

Fabricio Giusti Oliveira Monteiro

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Dual Citizenship - Brazil and Italy | <https://fabricio-giusti.github.io/Portfolio-Fabricio-Giusti/>

EDUCATION

Purdue University, College of Engineering, West Lafayette, IN

August 2024 – May 2028

Bachelor of Science in Mechanical Engineering

GPA: 3.89/4.0

Nanyang Technological University (NTU), Singapore

January 2027 – May 2027 (Expected)

Semester Abroad, Global Engineering Studies (GEARE Program)

PROFESSIONAL EXPERIENCE

EmbodimentX Inc.

San Mateo, CA

Robotics Engineer Intern

May 2026 – August 2026

- Deployed fine-tuned **Vision Language Action (VLA)** and **world action model (WAM)** policies onto dual UR3 arms and AgiBot G1/G2 humanoids in ROS 2, using asynchronous inference and a real-time chunking method.
- Fine-tuned **$\pi 0.5$** and world action models (ImageWAM, Fast-WAM) in PyTorch and JAX, reaching 70% to 90% success across seven bimanual manipulation tasks, adopted as the team's training baseline.
- Built the teleoperation data pipeline (Meta Quest VR) in ROS 2 and Python, recording 500+ demonstration episodes in LeRobot format and assembling the UR3 cell wiring, cameras, and lighting to remove visual artifacts in training data.
- Developed a self-hosted **Qwen2.5-VL** vision system for automated medicine label sorting and an offline voice-command launcher for live robot demos, and interviewed engineering candidates for the team.

RESEARCH & PROJECTS

Advanced Computation and Artificial Intelligence (ACAI) Lab

West Lafayette, IN

Undergraduate Research Assistant

January 2026 – Present

- Engineered four scripted bimanual policies for the 17-DOF Mobile ALOHA in **MuJoCo**, solving tube-picking and mid-air handovers with a Jacobian-based damped least-squares IK solver and grasp verification.
- Generated LeRobot datasets with dm_control domain randomization on NSF ACCESS HPC.

Nostradamus Project

West Lafayette, IN

Creator & Developer

September 2024 – December 2024

- Built a Python tool computing monthly hurricane probability for any city worldwide from 3M+ NOAA IBTrACS records since 1842, using vectorized pandas, OpenStreetMap geocoding API, and a Folium risk heatmap.

LEADERSHIP & COMMUNITY INVOLVEMENT

Boiler Soccer Robot Association (BSRA)

West Lafayette, IN

Co-Founder & President

November 2025 – Present

- Founded humanoid robotics club at Purdue to compete in the RoboCup Humanoid League, an international competition for fully autonomous multi-robot soccer teams.
- Led recruitment and secured necessary funding to launch the team, establishing clear development goals for the mechanical, electrical, and software subteams.

Engineering Projects in Community Service (EPICS) – Mobility (MOBI)

West Lafayette, IN

Design Lead

August 2024 – May 2025

- Directed a 5-member team for USA Blind Hockey, developing puck electronics in KiCad and LTspice while running Fusion 360 structural analysis on the polyurethane enclosure to validate 500 N impact durability with a hockey athlete.

BRASA – Brazilian Student Association at Purdue University

West Lafayette, IN

Workshop Director & Senior Advisory Board

September 2024 – Present

- Designed and delivered career workshops for 60+ attendees in a 200+ member community, a 50% increase from the prior year, directing a 4-member analyst team to shape topics from audience research and boost engagement.

AWARDS & CERTIFICATES

- Howard L. Timms Endowment Fund, Purdue University, School of Mechanical Engineering August 2026
- Outstanding Sophomore Award Finalist, Purdue University May 2026
- H. William Bottomley Research Scholarship, Purdue University January 2026
- Machine Learning Specialization, Stanford University August 2025

SKILLS

Software: Python, C++, C, MATLAB, Bash, Linux, Git, Docker, Google Cloud, Azure, REST APIs.

Robotics & ML: ROS 2, MuJoCo, LeRobot, PyTorch, JAX, OpenCV, CUDA, NumPy, Weights & Biases, openpi, Jetson.

Engineering Tools: Fusion 360, Siemens NX, Teamcenter, CATIA, KiCad, LTspice, 3DEXPERIENCE.

Languages: Portuguese (Native), English (Fluent), Spanish (Fluent), Italian (Basic), Mandarin (Basic).